

Assignment 2

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Impulse Response

1 Given the system $y(n) - y(n-1) - y(n-2) + x(n)$:

1.a Determine the (A, B, C, D) representation of the system.

1.a.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = y(n-2)$$

$$s_2(n) = y(n-1)$$

Then:

$$s_1(n+1) = y(n-1) = s_2(n)$$

$$s_2(n+1) =$$

$$= y(n)$$

$$= -y(n-2) + y(n-1) + x(n)$$

$$= -s_1(n) + s_2(n) + x(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$

1.a.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned} \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ -s_1(n) + s_2(n) + x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{A}$ and $\mathbf{B}$ matrices are $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively.

1.a.III Find $y(n)$

$$\begin{aligned} y(n) &= -y(n-2) + y(n-1) + x(n) \\ &= -s_1(n) + s_2(n) + x(n) \\ &= \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\overrightarrow{S(n+1)} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$y(n) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)$$

1.b Compute the impulse response $h(n)$

To compute the impulse response, we will use the state space-equations and solve for the input $x(n) = \delta(n)$.

Current State ($\overrightarrow{S(n)}$)

Current Output ($y(n)$)

$$\overrightarrow{S(0)} = \vec{0}$$

$$y(0) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$\overrightarrow{S(1)} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(0)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$y(1) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 1 \end{pmatrix} x(1) = 1$$

$$\overrightarrow{S(2)} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(1)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(2) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 1 \end{pmatrix} x(2) = 0$$

$$\overrightarrow{S(3)} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(2)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(3) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 1 \end{pmatrix} x(3) = 0$$

$$\overrightarrow{S(4)} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \overrightarrow{S(3)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(3) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(4) = \begin{pmatrix} -1 & 1 \end{pmatrix} \overrightarrow{S(4)} + \begin{pmatrix} 1 \end{pmatrix} x(4) = 0$$

$\vdots$

$\vdots$

Therefore, the impulse response of the system is given by $h(n) = \begin{cases} 1, & \text{if } n = 0, 1 \\ 0, & \text{otherwise} \end{cases} = \delta(n) + \delta(n-1)$.

Convolutional Systems

2 An LTI system is described by the impulse response $h(n) = \delta(n) + \delta(n - 2)$. Determine the output of the system for the input $\delta(n) + \delta(n - 2) + \delta(n - 4)$.

We can do this in 1 of two ways

2.I Solve using the LTI properties of the system

Since we know $h(n) = \delta(n) + \delta(n - 2)$, that implies that to output of the system $y(n) = \delta(n) + \delta(n - 2)$ when $x(n) = \delta(n)$. Therefore. Since the system is LTI (linear, time-invariant), we can deduce that:

$$\begin{aligned} y(n) &= h(n) + h(n - 2) + h(n - 4) \\ &= \delta(n) + \delta(n - 2) + \delta(n - 2) + \delta(n - 4) + \delta(n - 4) + \delta(n - 6) \\ &= \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6) \end{aligned}$$

$$\text{Thus, the output of the system is given by } y(n) = \begin{cases} 1 & \text{if } n = 0, 6 \\ 2, & \text{if } n = 2, 4 \\ 0, & \text{otherwise} \end{cases} = \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6).$$

2.II Solve using the convolutional equation

Using the convolutional equation, we can determine that:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} (h(n - k)x(k)) \\ &= \sum_{k=-\infty}^{\infty} ((\delta(n - k) + \delta(n - k - 2))(\delta(k) + \delta(k - 2) + \delta(k - 4))) \\ &= \sum_{k=-\infty}^{\infty} (\delta(n - k)\delta(k) + \delta(n - k)\delta(k - 2) + \delta(n - k)\delta(k - 4) \\ &\quad + \delta(n - k - 2)\delta(k) + \delta(n - k - 2)\delta(k - 2) + \delta(n - k - 2)\delta(k - 4)) \\ &= \delta(0)\delta(n) + \delta(0)\delta(n - 2) + \delta(0)\delta(n - 4) + \delta(0)\delta(n - 2) + \delta(0)\delta(n - 4) + \delta(0)\delta(n - 6) \\ &= \delta(n) + \delta(n - 2) + \delta(n - 4) + \delta(n - 2) + \delta(n - 4) + \delta(n - 6) \\ &= \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6) \end{aligned}$$

$$\text{Thus, the output of the system is given by } y(n) = \begin{cases} 1 & \text{if } n = 0, 6 \\ 2, & \text{if } n = 2, 4 \\ 0, & \text{otherwise} \end{cases} = \delta(n) + 2\delta(n - 2) + 2\delta(n - 4) + \delta(n - 6).$$

Convolution

3 Given the impulse response $h(n) = \begin{cases} 1, & \text{if } n = 0 \\ 2, & \text{if } n = 1 \\ 3, & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$, determine:

3.a The difference equation for the system

We know that when $x(n) = \delta(n)$, $y(n) = \begin{cases} 1, & \text{if } n = 0 \\ 2, & \text{if } n = 1 \\ 3, & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$. Therefore, we know:

$$y(n) = \delta(n) + 2\delta(n-1) + 3\delta(n-2)$$

Substituting in, using the fact $\delta(n) = x(n)$, we have:

$$y(n) = x(n) + 2x(n-1) + 3x(n-2)$$

Giving us the difference equation of our system.

3.b The (A, B, C, D) representation of the system

3.b.I Find $\overrightarrow{S(n)}$

To do this, we will use the difference equation for the system derives in a.).

Let:

$$s_1(n) = x(n-2)$$

$$s_2(n) = x(n-1)$$

Then:

$$s_1(n+1) = x(n-1) = s_2(n)$$

$$s_2(n+1) = x(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$

3.b.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned} \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{A}$ and $\mathbf{B}$ matrices are $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively.

3.b.III Find $y(n)$

$$\begin{aligned} y(n) &= 3x(n-2) + 2x(n-1) + x(n) \\ &= 3s_1(n-2) + 2s_2(n-1) + x(n) \\ &= \begin{pmatrix} 3 & 2 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} 3 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\overrightarrow{S(n+1)} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n)$$

$$y(n) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} x(n)$$

3.c The outputs of the system using convolutions for the input $x(n) = \delta(n) + \delta(n-1) + \delta(n-3)$

Using the convolutional equation, we can compute that the output of this system for the input $x(n) = \delta(n) + \delta(n-1) + \delta(n-3)$ is given by:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} (h(n-k)x(k)) \\ &= \sum_{k=-\infty}^{\infty} (h(n-k)(\delta(k) + \delta(k-1) + \delta(k-3))) \\ &= \sum_{k=-\infty}^{\infty} (h(n-k)\delta(k) + h(n-k)\delta(k-1) + h(n-k)\delta(k-3)) \\ &= h(n) + h(n-1) + h(n-3) \end{aligned}$$

And since we know $h(n) = \begin{cases} 1, & \text{if } n = 0 \\ 2, & \text{if } n = 1 \\ 3, & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases} = \delta(n) + 2\delta(n-1) + 3\delta(n-2)$, can conclude that:

$$\begin{aligned} y(n) &= h(n) + h(n-1) + h(n-3) \\ &= \delta(n) + 2\delta(n-1) + 3\delta(n-2) + \delta(n-1) + 2\delta(n-2) + 3\delta(n-3) + \delta(n-3) + 2\delta(n-4) + 3\delta(n-5) \\ &= \delta(n) + 3\delta(n-1) + 5\delta(n-2) + 4\delta(n-3) + 2\delta(n-4) + 3\delta(n-5) \end{aligned}$$

Therefore, the system's outputs for the input $x(n) = \delta(n) + \delta(n-1) + \delta(n-3)$ is given by:

$$y(n) = \begin{cases} 1, & \text{if } n = 0 \\ 3, & \text{if } n = 1 \\ 5, & \text{if } n = 2 \\ 4, & \text{if } n = 3 \\ 2, & \text{if } n = 4 \\ 3, & \text{if } n = 5 \\ 0, & \text{otherwise} \end{cases}$$

$$= \delta(n) + 3\delta(n-1) + 5\delta(n-2) + 4\delta(n-3) + 2\delta(n-4) + 3\delta(n-5)$$

Note here as well that we could have arrived at the conclusion that $y(n) = h(n) + h(n-1) + h(n-3)$ for the input $x(n) = \delta(n) + \delta(n-1) + \delta(n-3)$ without the use of the convolutional equation since, from a.), we know the system has no feedback and thus must be FIR.

Complex Numbers

4 Solve the following questions pertaining to complex numbers:

4.a Convert $-1 - i$ into exponential form

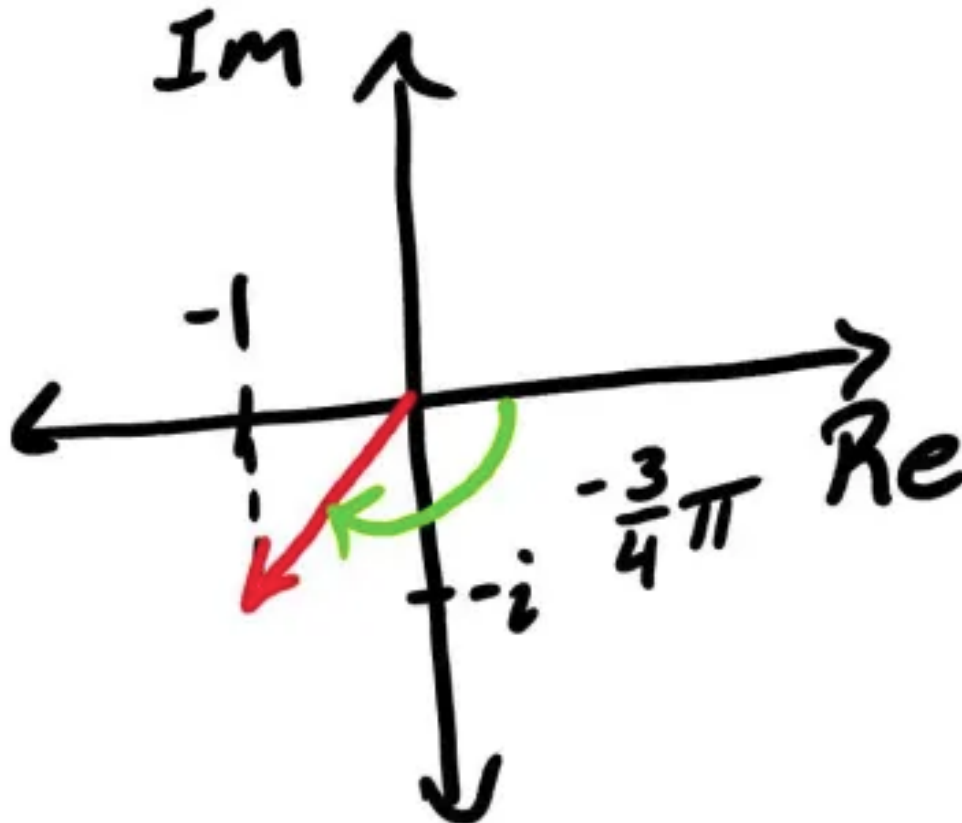


Figure 1: $-1 - i$ in the complex plane.

$$|-1 - i| = \sqrt{((-1)^2 + (-1)^2)} = \sqrt{2}$$

$$\arg(-1 - i) = \frac{-3\pi}{4}$$

Therefore, $-1 - i = \sqrt{2}e^{-\frac{3\pi}{4}i}$

4.b Compute $|e^{i\frac{\pi}{2}}|$

By the definition of complex exponentials, $|e^{\frac{\pi}{2}i}| = 1$.

4.c Convert -2 into exponential form

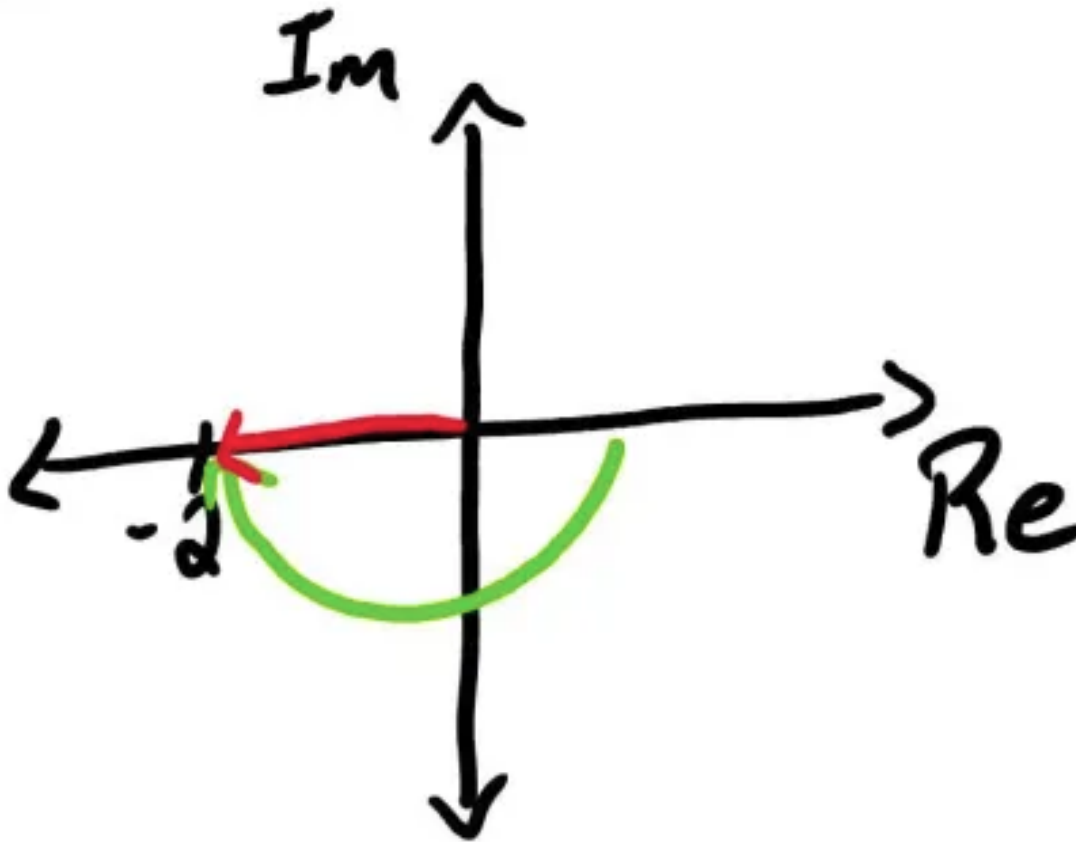


Figure 2: -2 in the complex plane.

$$\begin{aligned} |-2| &= 2 \\ \arg(-2) &= -\pi \end{aligned}$$

Therefore, $-2 = e^{-\pi i}$.

4.d Find $\arg(1 - i)$

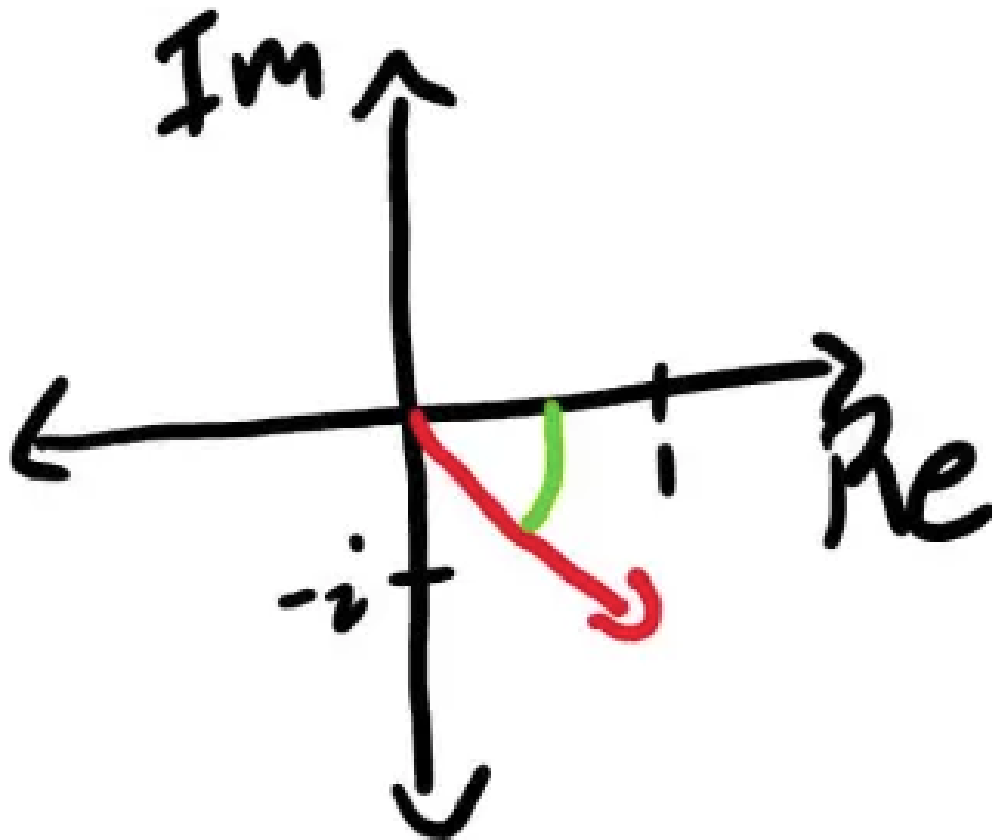


Figure 3: $1 - i$ in the complex plane.

Therefore, $\arg(1 - i) = \frac{-\pi}{4} \pm 2\pi k \quad k \in \mathbb{Z}$.

4.e Convert $-i$ into exponential form

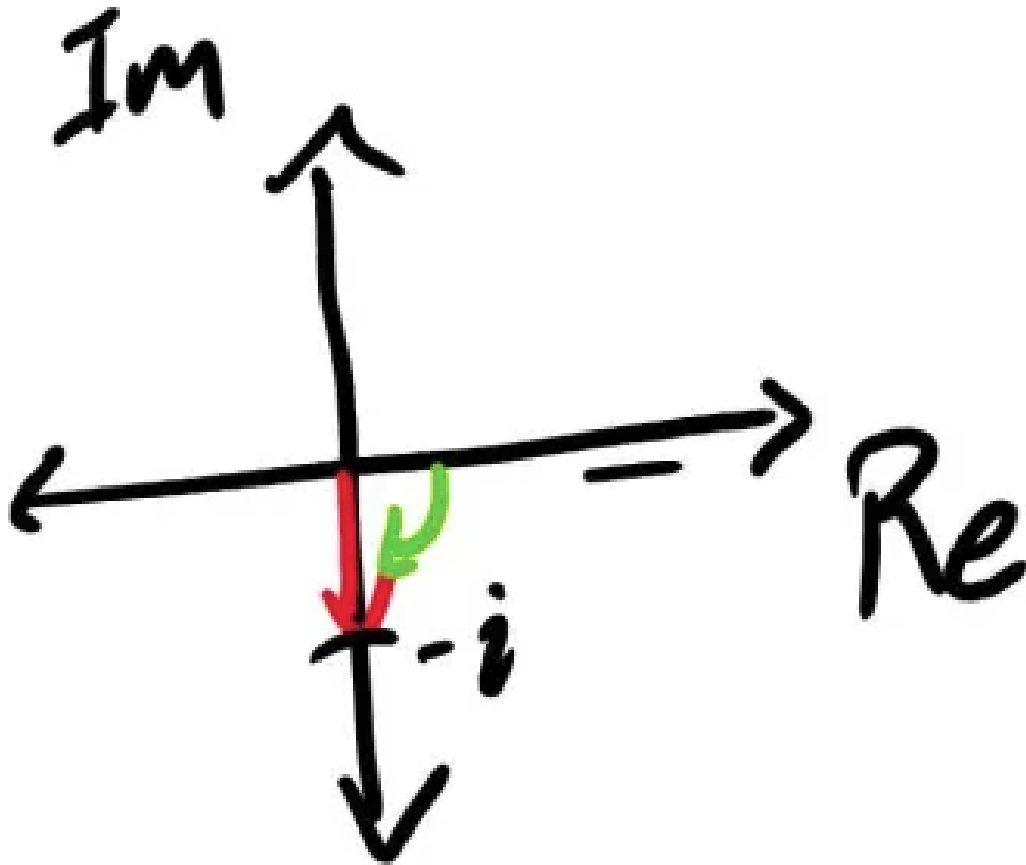


Figure 4: $-i$ in the complex plane.

$$|-i| = \sqrt{(-1)^2} = \sqrt{1} = 1$$
$$\arg(-i) = \frac{-\pi}{2}$$

Therefore, $-i = e^{-\frac{\pi}{2}i}$.